

Synthesis Report

Family: Cyclone IV

Device: GX EP4CGX150DF31I7AD

Optimization techniques used: Balanced (Normal Flow)

	Area	Dynami c Power	Max Frequency	LUTs	FFs	Memory	Setup Slack
Vanilla	12,897 LEs (9%)	25.69 mW	132.21 MHz	12,89 7	8,514	0 bit (but we have done a BRAM version, waiting to be updated)	1.038

Flow Summary Screenshot (See corelet-Flow Summary.rpt)

The screenshot displays the Quartus Prime IDE interface with the Flow Summary report open. The report is titled "Flow Summary" and shows the following details:

- Flow Status:** Successful - Sat Dec 13 00:36:52 2025
- Quartus Prime Version:** 25.1std.0 Build 1129 10/21/2025 SC Lite Edition
- Revision Name:** corelet
- Top-level Entity Name:** corelet
- Family:** Cyclone IV GX
- Device:** EP4CGX150DF31I7AD
- Timing Models:** Final
- Total logic elements:** 12,897 / 149,760 (9 %)
- Total registers:** 8514
- Total pins:** 427 / 508 (84 %)
- Total virtual pins:** 0
- Total memory bits:** 0 / 6,635,520 (0 %)
- Embedded Multiplier 9-bit elements:** 0 / 720 (0 %)
- Total GXB Receiver Channel PCS:** 0 / 8 (0 %)
- Total GXB Receiver Channel PMA:** 0 / 8 (0 %)
- Total GXB Transmitter Channel PCS:** 0 / 8 (0 %)
- Total GXB Transmitter Channel PMA:** 0 / 8 (0 %)
- Total PLLs:** 0 / 8 (0 %)

The report also includes a table of contents and a list of messages. The messages section shows the following:

- 223000 Starting Vectorless Power Activity Estimation
- 223001 Completed Vectorless Power Activity Estimation
- 218000 Using Advanced I/O Power to simulate I/O buffers with the specified board trace model
- 334003 Started post-fitting delay annotation
- 334004 Delay annotation completed successfully
- 215049 Average toggle rate for this design is 0.844 millions of transitions / sec
- 215031 Total thermal power estimate for the design is 320.36 mW
- Quartus Prime Power Analyzer was successful. 0 errors, 4 warnings

Max Frequency Screenshot (See corelet-Fmax Summary.rpt)

Slow 1200mV 100C Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	132.21 MHz	132.21 MHz	clk	

This panel reports FMAX for every clock in the design, regardless of the user-specified paths where the source and destination registers or ports are driven by the same clock. Generated clocks, are ignored. For paths between a clock and its inversion, FMAX is scaled along with FMAX, such that the duty cycle (in terms of a percentage) is maintained. Always use clock constraints and other clock reports for sign-off analysis.

Message Log:

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- 223001 Completed Vectorless Power Activity Estimation
- 218000 Using Advanced I/O Power to simulate I/O buffers with the specified board trace model
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- 215031 Total thermal power estimate for the design is 320.36 mW
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Power Analyzer Screenshot (See corelet-Power-Summary.rpt)

Power Analyzer Summary

Power Analyzer Status	Successful - Sat Dec 13 00:36:52 2025
Quartus Prime Version	25.1std.0 Build 1129 10/21/2025 SC Lite Edition
Revision Name	corelet
Top-level Entity Name	corelet
Family	Cyclone IV GX
Device	EP4CGX150DF31I7AD
Power Models	Final
Total Thermal Power Dissipation	320.36 mW
Core Dynamic Thermal Power Dissipation	25.69 mW
Core Static Thermal Power Dissipation	119.41 mW
I/O Thermal Power Dissipation	175.27 mW
Power Estimation Confidence	Low: user provided insufficient toggle rate data

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Timing Closure and Critical Path (See corelet-Slow 1200mV 100C Model .rpt)

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Slow 1200mV 100C Model report for corelet
Sat Dec 13 16:45:07 2025
Quartus Prime Version 25.1std.0 Build 1129 10/21/2025 SC Lite Edition

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; Table of Contents ;
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1. Legal Notice
2. Command Info
3. Summary of Paths
4. Path #1: Setup slack is 1.038
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: Command Info :
Report Timing: Found 1 setup paths (0 violated). Worst case slack is 1.038

Tcl Command:
report_timing -setup -multi_corner -panel_name {Report Timing} -apaths 1 -detail full_path

Options:
  -setup
  -apaths 1
  -detail full_path
  -panel_name {Report Timing}
  -multi_corner

Delay Model:
Slow 1200nm/ 100C Model

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Summary of Paths
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| Path ID | Start Node | End Node | Launch Clock | Launch Clock Relationship | Clock Skew | Data Delay |
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| 1.038 | mac_array_mac_array_inst[mac_row_row_mux[0]]_mac_row_instance[mac_tile_col_mux[1]]_mac_tile_instance[h_q[1]]_cifs0_cifs0_inst[cifs0_depath36_col_mux[0]]_cifs0_instance[q[6][15]]_clk | clk | 10.000 | -0.041 | 8.919 |
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Data Arrival Path									
Total	Incr	RF	Type	Fanout	Location	Element			
0.000	0.000					launch edge time			
3.384	3.384					clock path			
0.000	0.000					source latency			
0.000	0.000					clk			
0.000	0.000					clk-input[i]			
0.702	0.702					clk-input[i]			
0.000	0.000	RR	IC	1	PIN N15	101BFF_XST_Y0_N15			
0.000	0.000	RR	IC	1	PIN N15	101BFF_XST_Y0_N15			
1.093	0.000	RR	IC	8514	CLKCTRL_049	clk-input[clkctrl0:clkctrl0]			
2.783	0.690	RR	IC	1	FF_XF0_Y25_N1	mac_array_inst[row_num[8]:mac_row_instance[col_num[]:mac_tile_instance[b_q[]]]:clk			
4.801	0.601	RR	IC	1	FF_XF0_Y25_N1	mac_array_inst[row_num[8]:mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
12.303	8.601					data path			
3.616	0.232					mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
3.616	0.000	FF	CELL	16	FF_XF0_Y25_N1	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.615	0.000	FF	CELL	16	FF_XF0_Y25_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
4.663	0.428	FF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.587	0.394	FF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.581	0.000	FF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.282	0.067	RF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.282	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.346	0.428	FF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.446	0.276	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.446	0.004	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.446	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
6.385	0.139	RR	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.937	0.652	RR	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.916	0.479	RR	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.916	0.000	FF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.067	RF	CELL	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8	mac_array_inst[mac_row_instance[col_num[8]:mac_row_instance[mac_tile_instance[b_q[]]]:mac_tile_instance[b_q[]]]			
0.983	0.000	FF	IC	2	LCOONE_X0_Y0_N8				